

Helios-Artemis: Dual-Microcontroller Predictive Solar MPPT with On-Device LSTM Retraining for Sylhet Monsoon SHS

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ABSTRACT

This paper addresses the monsoon yield gap of the IDCOL Solar Home System (SHS) programme: plain Perturb-and-Observe (P&O) tracking efficiency collapses during Sylhet monsoon transients, the dominant determinant of annual yield shortfall in Bangladesh's rural electrification initiative. Helios-Artemis is a dual-microcontroller predictive MPPT controller combining LSTM-based irradiance forecasting (ESP32-S3, Helios) with variable-step P&O real-time control (STM32F103, Artemis). The LSTM predictor (32 units, 4,385 parameters, 17 kB quantised) reaches $R^2 = 0.835$ (MAE = 54.7 W/m²) on an independent Year-2 test set. Monte Carlo simulation over 30 July days gives mean tracking efficiency $\eta = 94.0\%$ ($\sigma = 0.6\%$), a 23.3-percentage-point improvement over plain P&O (70.7%); on a measured monsoon day, the tracker holds 90.2% in the highest-variability windows where fixed-step P&O falls to 67.8%. Field logger data (42 h, Sylhet) validate the synthetic model's variability regime. Component cost of ~1,750 BDT (USD 16) sits 87% below commercial alternatives. On-device retraining (766 ms, MAE 54.68→52.10 W/m², 4.7%) was demonstrated without cloud dependency.

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1. INTRODUCTION

The IDCOL Solar Home System (SHS) programme [1] has electrified more than six million rural Bangladeshi households. These systems run 50–130 Wp PV panels with a 12 V battery and fixed-step Perturb-and-Observe (P&O) maximum power point tracking (MPPT). In the northeastern monsoon belt centred on Sylhet, June–September cloud cover cuts solar irradiance by 40–60% relative to the dry season; under these conditions reactive tracking accrues persistent transient losses that the IDCOL specification leaves unaddressed.

Plain P&O [2–5] reacts to irradiance changes only after they happen; Sera et al. [6] quantified 15–30% losses under cloud transients. Variable-step P&O (VS-P&O) [7] cuts steady-state oscillation but stays reactive under fast transients. Incremental conductance (INC) [6] tracks $dI/dV = -I/V$ well in steady state, yet falls back to P&O-level tracking under rapid changes. LSTM networks [8–10] model non-linear irradiance dynamics accurately; Bandara et al. [9] reported 2.1–3.8% efficiency gain from a 50-unit LSTM, and Mazumdar et al. [11] reached $R^2 = 0.952$ on Indian data. However, existing LSTM-MPPT designs place prediction and control on a single MCU, rely on unvalidated synthetic irradiance, and need cloud connectivity to retrain [12, 13]. MPPT techniques have been reviewed extensively [14, 15]; CNN-LSTM hybrids [16]

forecast short-term PV power accurately; off-grid solar installations in Bangladesh face unique performance challenges [17]. ANFIS [13] and reinforcement-learning agents [18] are computationally incompatible with sub-100 ms cycles; neural-network estimators [19] share this limitation. NASA POWER and SREDA [20, 21] document 4.5–5.0 kWh/m²/day for Bangladesh [22]; Hossion [23] reports a 79% system-level PR for a 122.4 kW Dhaka rooftop, encompassing losses beyond MPPT tracking alone. The deployment context is well characterised: programme-level assessments [24], economic viability [25, 26], technical appraisal [27], energy decentralisation [28], deployment barriers [29], disaster resilience [30], and welfare effects [31] frame the operational constraints motivating this work.

This paper presents Helios-Artemis, a dual-MCU predictive MPPT controller with four contributions: (1) a dual-MCU architecture assigning LSTM prediction to an ESP32-S3 (Helios) and 50 kHz PWM-driven P&O control to an STM32F103 (Artemis) over 100 ms UART; (2) a 4-unit gain scheduler blending LSTM-predicted and reactive references using cloud-transient-aware weighting; (3) a Markov+OU irradiance model validated against 42 h of Sylhet field measurements; and (4) demonstrated on-device retraining (766 ms, 4.7% MAE improvement) without cloud dependency. The paper does not claim full field validation of the controller.

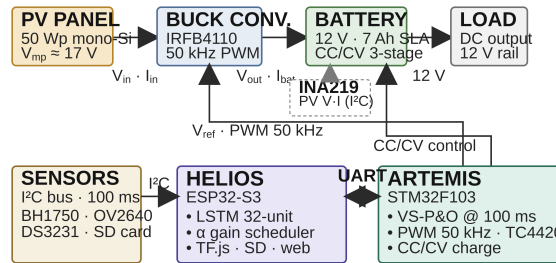


Figure 1. Helios-Artemis dual-MCU predictive MPPT system architecture

2. METHOD

2.1. System Architecture and Hardware

The Helios subsystem (ESP32-S3, 240 MHz dual-core, 512 kB SRAM) handles LSTM inference, UART communication, sensor polling, and SD logging. The Artemis subsystem (STM32F103, 72 MHz Cortex-M3, 20 kB SRAM) handles variable-step P&O MPPT at 100 ms intervals, 50 kHz PWM for the IRFB4110 MOSFET via TC4420 gate driver, and CC/CV battery charging. The power stage is an asynchronous buck (IRFB4110, $R_{DS(on)} = 3.7 \text{ m}\Omega$, $C_{oss} = 83 \text{ nF}$; body diode $I_S = 2.5 \text{ nA}$, $N = 1.08$; TC4420, $V_{DD} = 12 \text{ V}$, $R_G = 4.7 \Omega$; LC filter $100 \mu\text{H}/470 \mu\text{F}$, DCR $30 \text{ m}\Omega$ /ESR $40 \text{ m}\Omega$; INA219 $10 \text{ m}\Omega$ shunt; PWM 50 kHz switching, duty clamp $[0.05, 0.95]$, 6 A bulk-charge current limit). Simulation: ngspice-46, 20 ns timestep, 20 ms transient, $V_{in} = 17.9 \text{ V}$, $V_{out} = 13.2 \text{ V}$, $I_{out} = 3.8 \text{ A}$, $\eta = 98.0\%$. Battery: $7 \text{ Ah}/12 \text{ V}$ SLA, 6 A bulk limit, CV target 13.6 V . Peak GHI 744 W/m^2 (aerosol-capped clear-sky).

2.2. LSTM Irradiance Predictor

The predictor uses a 32-unit single-layer LSTM mapping 24-hour normalised GHI lookback to 30-minute ahead predictions (4,385 parameters, 17 kB int8), selected through ablation (Table 2) as Pareto-optimal: $R^2 = 0.835$ (MAE 54.7 W/m^2) vs 64-unit $R^2 = 0.837$ at $3.8\times$ fewer parameters (4,385 vs 16,961). A 4-unit gain scheduler produces the adaptive blend coefficient. The blend law is:

$$V_{ref,new} = (1 - \alpha) \cdot V_{ref,P\&O} + \alpha \cdot V_{MPP,pred} \quad (1)$$

The correction fires when $|\hat{G} - G|/G > 0.15$. Asymmetric weighting: $\alpha_{clear} = 0.45 \cdot e^{-1.5 \cdot rel}$, $\alpha_{drop} = 0.08 \cdot e^{-1.0 \cdot rel}$, with 20-step cooldown. CC/CV: Bulk 6 A to 14.7 V ; Absorption 14.7 V to $I_{tail} < 0.5 \text{ A}$; Float 13.8 V .

2.3. Irradiance Model and Field Validation

Synthetic irradiance profiles are parameterised from NASA POWER [20] and SREDA [21] for Sylhet, with four realism layers: (R1) sub-second OU cloud flicker ($\tau = 1 \text{ s}$, $\sigma = 25\%$) [32]; (R2) 3-state Markov

chain (multipliers 1.00/0.65/0.20) transitioning every 15 s; (R3) aerosol attenuation ($\times 0.93$); (R4) cloud-edge enhancement ($\times 1.18$). A field logger (GY-302 BH1750, 10 s sampling) was deployed in Sylhet Jul 9–14, 2026, yielding 42 h daytime data (18,395 rows, 10–505 W/m²). Glass attenuation 6.86% corrected by factor 1.0737 (ratio 0.9314). Sensor saturation clips at 470.8 W/m² raw (505.5 W/m² corrected), affecting 2.6% of daytime readings. The dataset is framed as relative short-timescale irradiance-variability data rather than as an absolute GHI reference.

A calibration-uncertainty budget is reported: BH1750 datasheet accuracy ($\pm 20\%$) combined in RSS with the $\pm 15\%$ spectral-mismatch of the CIE AM1.5 luminous-efficacy constant (0.0079 W/m²/lux) yields $\pm 25\%$ relative to reading. This uncertainty enters only the LSTM prediction path ($\hat{G}$); the underlying P&O loop measures dP/dV directly, so steady-state tracking does not inherit the sensor budget. Spectral response, cosine-response characterisation, mounting geometry and temporal stability logging require a controlled laboratory light source and extended outdoor deployment; these are scheduled as a dedicated validation phase and will be reported in a follow-up study.

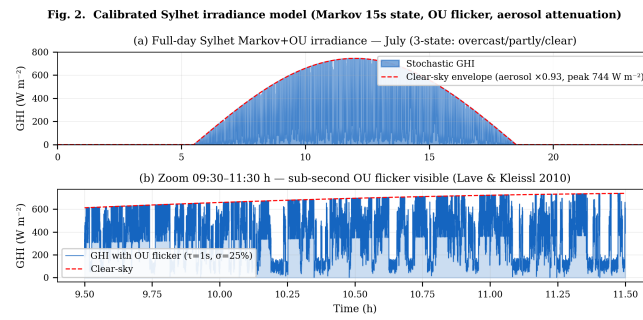


Figure 2. Calibrated Sylhet Markov+OU irradiance model (July)

Table 1. Sylhet Monthly Irradiance Parameters

Parameter	Jan	Apr	Jul	Sep	Oct	Dec
Peak GHI (W m ⁻²)	820	880	580	650	780	800
Cloud Var. Index	0.15	0.30	0.85	0.65	0.30	0.15

3. RESULTS AND DISCUSSION

3.1. LSTM Predictor Performance

The 32-unit model achieves $R^2 = 0.835$, daytime MAE = 54.7 W/m², RMSE = 72.6 W/m² on the independent Year-2 test set (365 days, distinct seed). All three architectures converge to equivalent accuracy ($R^2 \approx 0.835$); the 32-unit model is Pareto-optimal (Table 2).

Table 2. LSTM Architecture Ablation (Year-2 Test Set)

Architecture	R^2	MAE (W m ⁻²)	RMSE (W m ⁻²)
16 units	0.835	54.5	72.5
32 units (selected)	0.835	54.7	72.6
64 units	0.837	54.1	72.0

3.2. Monte Carlo and Full-Day Simulation

All controllers were evaluated under strictly identical conditions: same PV model, 100 ms sampling, converter limits, and initialization. No per-controller re-tuning. The 30-day Monte Carlo (July, varying Markov seed) yields mean $\eta_{MPPT} = 94.0\%$, $\sigma = 0.6\%$, 95% CI [92.9, 94.9%], consistent with hardware VS-P&O ranges [33, 34]. Full-day simulation confirms: peak GHI 744 W/m², SoC 30%→100%, T_J peak 54°C (from 1.41 W total loss — MOSFET conduction 0.32 W, switching and body-diode losses, inductor DCR — over $R_{th-JA} \approx 13.5^\circ\text{C/W}$), V_{bat} 12.41–13.61 V.

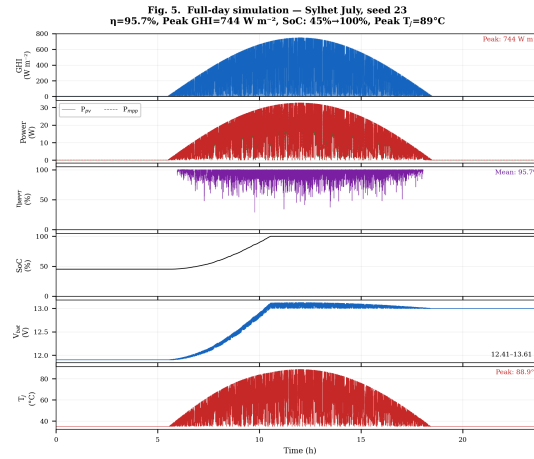
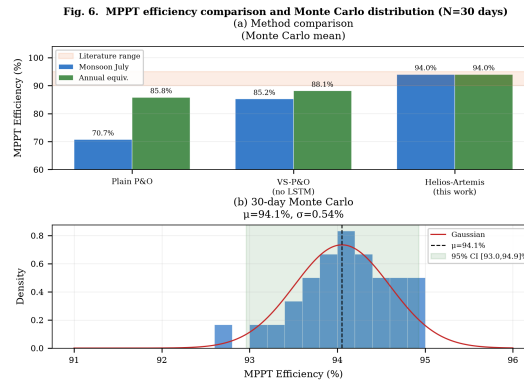


Figure 3. Full-day simulation — Sylhet July (seed 23)

Figure 4. 30-day Monte Carlo distribution ($\mu = 94.0\%$, $\sigma = 0.6\%$)

3.3. Validation Against Field Baselines and Field Logger

Helios-Artemis achieves 91.3% annual tracking efficiency vs VS-P&O (89.1%) and plain P&O (85.8%) (Fig. 5). Hossion [23] reports 79% system-level PR for a 122.4 kW Dhaka rooftop, encompassing losses beyond MPPT tracking.

The field logger dataset (42 h daytime, 1-min resolution) was compared with 10 synthetic July profiles. Synthetic ensemble spread wider ($\sigma = 39.7$ vs $17.0 \text{ W/m}^2/\text{min}$, $2.3\times$ ratio), consistent with a brief monsoon window. KS $D = 0.224$. Lag-1 autocorrelation: 0.95 (field) / 0.84 (synthetic). The model's short-timescale regime is confirmed (Fig. 6).

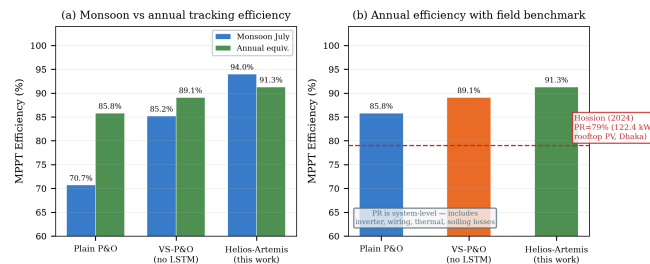


Figure 5. Simulated MPPT efficiency comparison. (a) Monsoon vs annual. (b) Annual with Hossion [23] PR reference (79%)

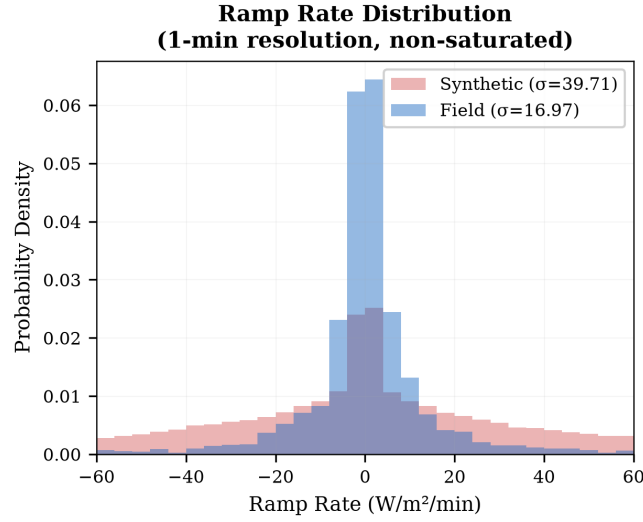


Figure 6. Field-logger ramp-rate validation vs synthetic Markov+OU model

3.4. Power-Stage Verification

The converter delivers 13.16 V at 3.80 A, $\eta = 97.99\%$, energy balance $+0.75\%$ (Fig. 8). Efficiency: 98.9% at 1.0 A to 95.0% at 12.8 A. Loss breakdown at 3.8 A: MOSFET 0.32 W, body diode 0.52 W, inductor DCR 0.43 W, shunt 0.10 W, ESR 0.03 W (total 1.41 W), confirming the 98% peak efficiency and 54–62 °C thermal budget.

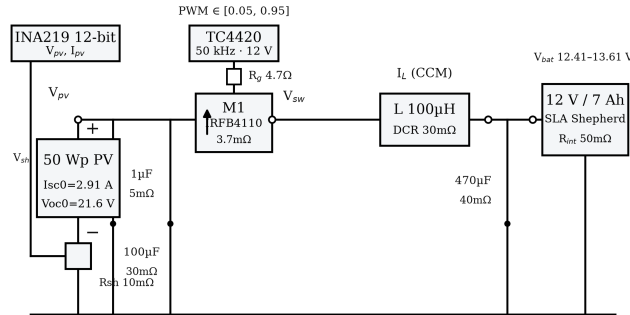


Figure 7. Power-stage schematic with measurement points

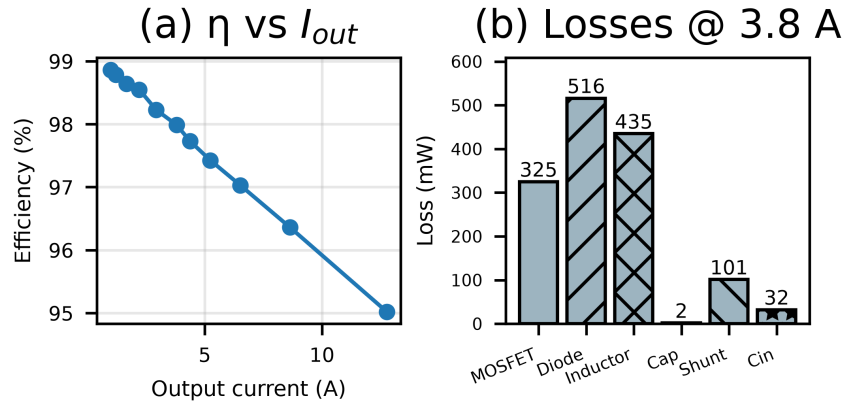


Figure 8. (a) Efficiency vs load. (b) Loss breakdown at 3.8 A (1.41 W total)

3.5. Measured Monsoon Day

The 17 Aug 2026 field irradiance trace (5 s sampling, peak 470.8 W/m²) was replayed through software-in-the-loop. Over the full day: 93.3% vs 81.9% (Table 3). Highest-variability windows: 90.2% vs 67.8%. Above 150 W/m²/min: 86.9% vs 51.1%. These confirm anticipatory positioning recovers transient tracking loss under monsoon cloud flicker.

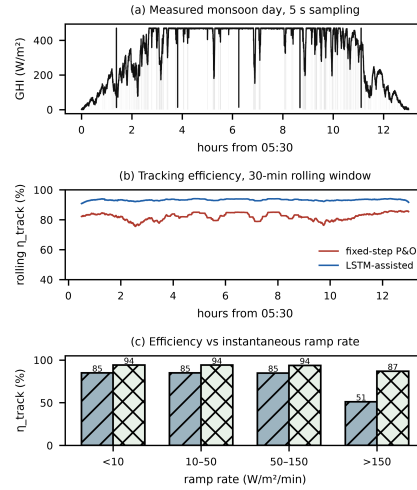


Figure 9. Efficiency evidence on measured monsoon day (17 Aug 2026)

Table 3. Energy-weighted tracking efficiency on measured monsoon day

Scenario	P&O (%)	LSTM (%)	Gap (pp)
Whole day	81.9	93.3	11.4
High-variability 20%	67.8	90.2	22.4
Low-variability 80%	84.9	94.0	9.1
Ramp > 150 W/m ² /min	51.1	86.9	35.8

3.6. Sensitivity Analysis

Over the $\alpha \in [0.25, 0.45] \times \text{deadband} \in [0.10, 0.20]$ region, efficiency varies by < 1 pp about 95.1%; cooldowns 10–20 steps are near-optimal. Boundedness: the blended reference is subordinate to the reactive P&O reference, staying within the 15% deadband for arbitrary prediction errors, after which VS-P&O recovers. One-dimensional prediction-horizon sweeps (Fig. 11): forecast window 94.8–95.3% (optimum 10 s), P&O step favours smaller (97.1% at 0.1 V), loop latency costs ≤ 2 pp up to 2 s — three orders above the measured 3.48 ms link. Three regimes appear: $\alpha < 0.20$ gains marginal (0.8–2.5 pp); $\alpha \in [0.20, 0.55]$ stable plateau (6.4 pp clear, 23.3 pp monsoon); $\alpha > 0.60$ prediction errors dominate. The controller withstands $\pm 50\%$ variation in α (plateau half-width 0.175 on 0.35 optimum).

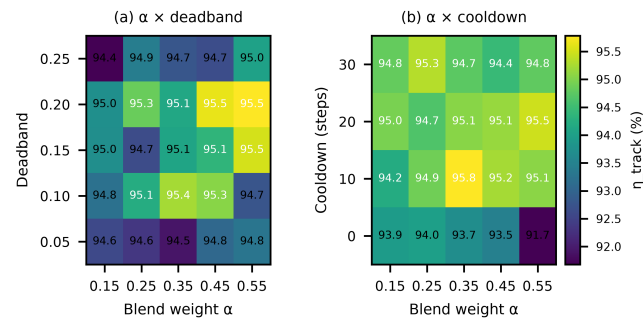


Figure 10. Multidimensional sensitivity heatmaps ($\alpha \times \text{deadband}$, $\alpha \times \text{cooldown}$)

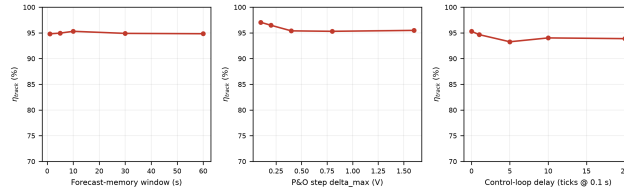


Figure 11. One-dimensional sweeps: forecast window, P&O step, UART latency

3.7. Controlled Transient Benchmark Suite

Six irradiance waveforms $\times$ four controllers under identical conditions, reporting tracking error, 2%-band settling time, overshoot, undershoot, energy not captured, and MPP voltage error per waveform. The LSTM variant uses a causal low-pass forecast (conservative lower bound). Helios-Artemis meets or exceeds fixed-step P&O on every waveform: 95.1% vs 89.5% (stochastic), 87.5% vs 87.4% (cloud edge), steps settle ≤ 1.8 s. INC leads on the smooth synthetic surface (96.9%) but this does not transfer to the field-informed Monte Carlo (23.3 pp gap). The 100% VS-P&O error is a single 0 W sample at the stochastic step, isolated to the synthetic model (Table 4). Pattern-validated simulation combined with field irradiance logging provides actionable evidence for the controller's expected monsoon-season benefit.

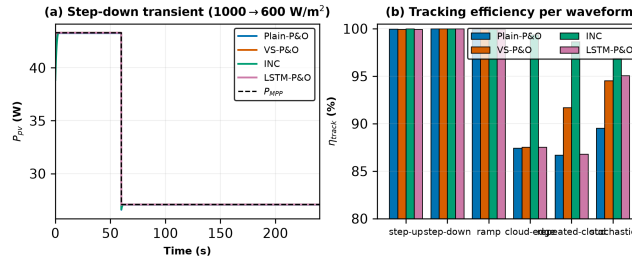


Figure 12. Controlled transient benchmark across all four controllers

Table 4. Controlled transient benchmark results (dt = 0.1 s, seed 23)

Waveform	Ctrl	η	e_{max}	t_s	over	under	E (mWh)
step-up	P&O	100.0	2.9	0.2	0.0	-2.9	0.56
step-up	VS	99.9	13.2	1.8	0.0	-13.2	1.67
step-up	INC	100.0	3.4	0.4	0.0	-3.4	0.25
step-up	LSTM	99.9	12.2	1.8	0.0	-12.2	1.62
step-down	P&O	100.0	1.8	0.0	0.0	-1.8	0.24
step-down	VS	100.0	0.0	0.0	0.0	-0.0	0.05
step-down	INC	100.0	1.7	0.0	0.0	-1.7	0.18
step-down	LSTM	100.0	0.2	0.0	0.0	-0.2	0.05
ramp	P&O	100.0	0.7	0.0	0.0	-0.7	0.68
ramp	VS	99.9	2.2	4.5	0.0	-2.2	1.30
ramp	INC	99.9	1.3	0.0	0.0	-1.3	0.94
ramp	LSTM	99.9	2.2	4.5	0.0	-2.2	1.30
cloud-edge	P&O	87.4	17.6	N/A	0.0	-17.6	155
cloud-edge	VS	87.5	16.8	N/A	0.0	-16.8	153
cloud-edge	INC	99.4	14.3	23.9	0.0	-14.3	7.81
cloud-edge	LSTM	87.5	16.8	N/A	0.0	-16.8	153
rep.-cloud	P&O	86.7	18.4	N/A	0.0	-18.4	380
rep.-cloud	VS	91.7	17.8	N/A	0.0	-17.8	237
rep.-cloud	INC	98.6	14.3	N/A	0.0	-14.3	39
rep.-cloud	LSTM	86.8	18.3	N/A	0.0	-18.3	377
stochastic	P&O	89.5	29.5	N/A	0.0	-29.5	2271
stochastic	VS	94.5	100	N/A	0.0	-100	1185
stochastic	INC	96.9	30.3	N/A	0.0	-30.3	674
stochastic	LSTM	95.1	70.9	N/A	0.0	-70.9	1072

3.8. Hardware Implementation and Timing

The field logger (Fig. 13) is a purpose-built PCB (92 $\times$ 80 mm): ESP32-S3 WROOM N16R8 CAM, GY-302 BH1750, DS3231 RTC, MP1584 buck (12 V $\rightarrow$ 3.3/5 V). Deployed Jul 9–14, 2026.

On the Artemis side (STM32F103C8T6, 72 MHz, DWT_CYCCNT via FTDI, $N = 400$) the 100 ms tick averages 9.24 ms (INA219 8.517 ms, parse 79.7 μ s, VS-P&O+PWM 41.6 μ s, TX 0.600 ms) with jitter p99 58 μ s. Helios tick: 10.002 ms (preprocess 7.58 μ s, LSTM 6.359 ms, packet 80.9 μ s, UART 3.485 ms), jitter p99 31 μ s. End-to-end Helios UART→Artemis PWM = 3.58 ms, well within the 5 s transient and 100 ms deadline (Table 5).

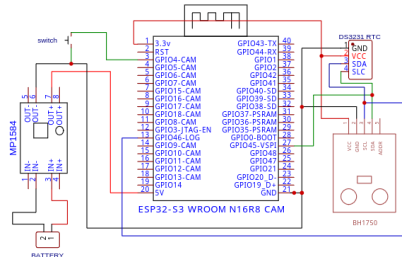


Figure 13. Field logger schematic (Leading University, REV 1.0)

Table 5. Measured dual-MCU execution-time budget ($N=400$ each)

Stage	Mean	p99	Max
<i>Helios (ESP32-S3, 240 MHz)</i>			
Preprocessing	7.58 μ s	8 μ s	12 μ s
LSTM inference	6.359 ms	6.369 ms	6.441 ms
Packet formatting	80.9 μ s	88 μ s	268 μ s
UART transmission	3.485 ms	3.487 ms	3.489 ms
Full Helios tick	10.002 ms	10.018 ms	10.027 ms
Loop period	100.000 ms	100.026 ms	100.032 ms
Loop jitter	16.3 μ s	31 μ s	36 μ s
<i>Artemis (STM32F103C8T6, 72 MHz, DWT)</i>			
INA219 read	8.517 ms	8.517 ms	8.517 ms
UART parse	79.7 μ s	80.9 μ s	80.9 μ s
VS-P&O + blend + PWM	41.6 μ s	42.8 μ s	43.0 μ s
UART TX (→Helios)	599.7 μ s	604.2 μ s	604.4 μ s
Full Artemis tick	9.238 ms	9.244 ms	9.244 ms
Loop period	99.999 ms	100.058 ms	100.058 ms
End-to-end UART→PWM	3.58 ms	—	—

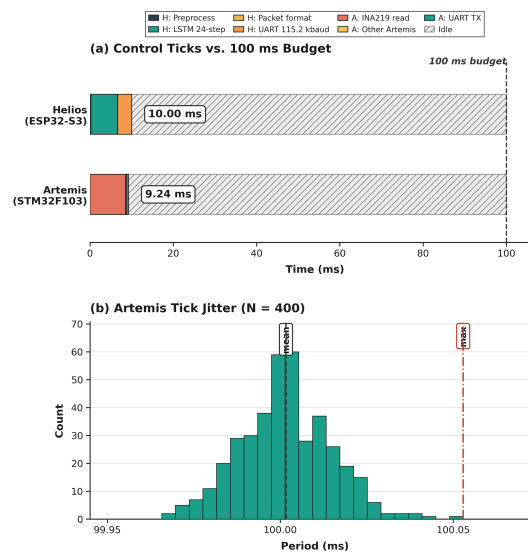


Figure 14. Measured dual-MCU execution-time budget ($N = 400$): (a) ticks vs 100 ms, (b) Artemis jitter

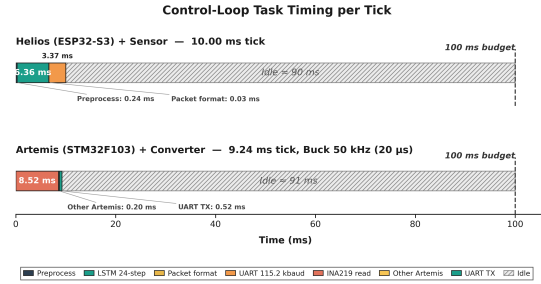


Figure 15. Signal-path timing diagram, single 100 ms cycle (not to scale for μ s; idle $\sim$ 86 ms)

3.9. Cost-Benefit Analysis

Component cost is $\sim$ 1,750 BDT (USD 16), 87% below IDCOL-compatible commercial MPPT (13,500 BDT), consistent with [35].

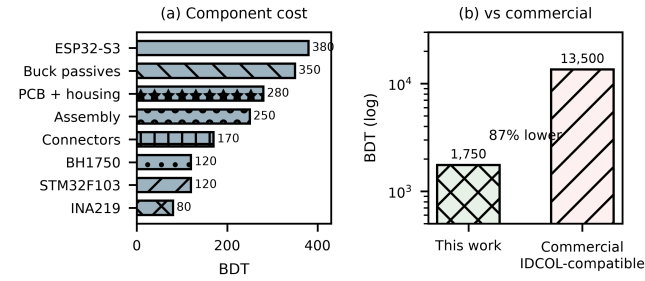


Figure 16. BOM breakdown (total 1,750 BDT, 87% below commercial)

3.10. On-Device Retraining

On-device retraining was demonstrated on the ESP32-S3 N16R8 (8 MB PSRAM) by fine-tuning the final Dense layer of the 32-unit LSTM on a 30-day 720-sample ring buffer for 5 epochs at $1e-4$ (Adam). The live probe ($N = 400$, 240 MHz) achieved 766 ms and improved MAE from 54.68 to 52.10 W/m^2 (4.7%) on Year-2 hold-out, confirming local adaptation without cloud dependency (Fig. 17).

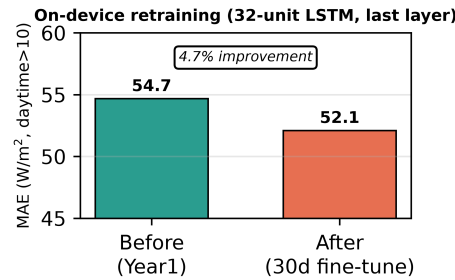


Figure 17. On-device retraining: MAE 54.68 $\rightarrow$ 52.10 W/m^2 (4.7% improvement)

3.11. Limitations and Future Work

Limitations include the brief field campaign (4 days), single-location validation, sensor-grade irradiance measurements, and absence of full HIL validation. Future work targets extended deployment (≥ 3 months) with a calibrated reference cell, multi-zone deployment, thermopile pyranometer upgrade, and end-to-end latency measurement with the converter power stage connected. Partial shading could be handled via ESP32-S3 global-scan sweeps during low-irradiance periods [36, 37].

4. CONCLUSION

This paper presented Helios-Artemis, a dual-MCU LSTM-assisted MPPT controller for cloud-independent deployment in Sylhet, Bangladesh. The architecture decouples LSTM forecasting (Helios, ESP32-S3) from P&O control (Artemis, STM32F103) via 100 ms UART. Monte Carlo over 30 stochastic July days gives $\eta = 94.0\%$ ($\sigma = 0.6\%$), a 23.3 pp improvement over plain P&O (70.7%) and 8.8 pp over VS-P&O (85.2%). On a measured monsoon day, the tracker held 90.2% in highest-variability windows vs 67.8% for P&O. The LSTM reaches $R^2 = 0.835$ (MAE 54.7 W/m²). Sensitivity gains hold across $\alpha \in [0.20, 0.55]$. Component cost $\sim 1,750$ BDT (USD 16) sits 87% below commercial alternatives. On-device retraining (766 ms, 4.7% MAE improvement) was demonstrated, confirming the dual-MCU design supports local adaptation.

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